

Data Collection Techniques

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Data are facts, numbers, or observations gathered for a specific purpose.

Data Collection

Data collection is the process of gathering information from people, objects, or events.

Different **data collection techniques** are used depending on the type of information needed.

Survey

A **survey** gathers information by asking questions through questionnaires or interviews.

For example, a teacher may ask students about their favorite school subject.

Observation

An **observation** involves carefully watching and recording events or behaviors.

For example, a teacher may record how many students arrive late every morning.

Experiment

An **experiment** is used to test a hypothesis by changing one factor and observing the results. For example, a scientist may change the amount of water given to plants and observe their growth.

Measurement

A **measurement** collects numerical data such as length, mass, time, or temperature.

For example, a tailor may determine the waist sizes of customers.

Census and Sample Survey

A **census** gathers information from every member of a population.

A **sample survey** collects data from only a part of the population.

Choosing a Technique

Choosing the appropriate data collection technique helps ensure that the collected data are accurate and useful.

The technique should match the type of information needed.

Examples

Example 1

What is the favorite school subject of all Grade 7 students?

The appropriate technique is a

Survey

because information is gathered by asking questions.

Example 2

A scientist wants to know how the growth of a plant changes when different amounts of water are given.

The appropriate technique is an

Experiment

because a factor is changed and the results are observed.

Example 3

A teacher records how many students arrive late every morning for two weeks.

The appropriate technique is

Observation

because the behavior is watched and recorded.

Example 4

A tailor determines the waist sizes of customers.

The appropriate technique is

Measurement

because numerical data are collected.

Example 5

The school principal wants information from every student regarding internet access at home.

The appropriate method is a

Census

because every member of the population is included.

Practice Exercises

Practice 1

Identify the most appropriate data collection technique.

“What is the favorite school subject of all Grade 7 students?”

Practice 2

Identify the most appropriate data collection technique.

“A scientist wants to know how the growth of a plant changes when different amounts of water are given.”

Practice 3

Identify the most appropriate data collection technique.

“A teacher records how many students arrive late every morning for two weeks.”

Practice 4

Identify the most appropriate data collection technique.

“A tailor determines the waist sizes of customers.”

Practice 5

Identify the most appropriate data collection technique.

“The school principal wants information from every student regarding internet access at home.”

Practice 6

Tell whether each situation uses a **census** or a **sample survey**.

A researcher asks all 520 Grade 7 students about their favorite sport.

Practice 7

Tell whether each situation uses a **census** or a **sample survey**.

A researcher interviews only 80 randomly selected Grade 7 students.

Practice 8

Match the situation with the correct technique.

(A) Survey

(B) Observation

(C) Experiment

(D) Measurement

“A nurse records the height of 25 children.”

Practice 9

Which technique is best for determining the number of vehicles passing through a school gate between 7:00 a.m. and 8:00 a.m.?

Activities

Activity 1

Identify the most appropriate data collection technique.

“What is the favorite snack of all Grade 7 students?”

Activity 2

Identify the most appropriate data collection technique.

“A student investigates how sunlight affects the growth of bean plants.”

Activity 3

Identify the most appropriate data collection technique.

“A class officer records how many students borrow books from the library each day.”

Activity 4

Identify the most appropriate data collection technique.

“A carpenter determines the lengths of wooden boards.”

Activity 5

Identify the most appropriate data collection technique.

“A barangay captain gathers information from every household about waste disposal.”

Activity 6

Tell whether each situation uses a **census** or a **sample survey**.

A school asks every teacher to complete a satisfaction questionnaire.

Activity 7

Tell whether each situation uses a **census** or a **sample survey**.

A school interviews only 40 parents selected at random.

Activity 8

Match the situation with the correct technique.

(A) Survey

(B) Observation

(C) Experiment

(D) Measurement

“A doctor records the body temperature of patients.”

Activity 9

Which technique is best for determining how many students enter the canteen during recess?

Thank you!